Acer had acceptable black levels, but got oversaturated when the intensity was increased. The BenQ wasn't very keen on proper black reproduction and greyscales. All of them had varying levels of colour shift, with the problem being severe with the Epson. Every DLP projector except the Sharp had problems rendering the colour gradations and the blacks, and the subsequent greyscales were plagued with impurities ranging from bluish to red from one projector to another. Even the LCDs weren't spared, with the Hitachi giving better results when compared to the pathetic performance of the Epson.

The projectors with phase problems fared poorly in the sharpness and resolution tests. The Epson's performance in this segment was at rock-bottom levels, because its inherent problems prevented it from rendering any of the test patterns properly. The Sharp was the absolute winner in this segment, followed by the BenQ. The Acer and Epson had problems with focusing and resolution detail at the corners of the screen. The BenQ, due to its lower phase issues, scored very well in the moiré pattern test, while Sharp once again aced the segment with virtually non-existent moiré patterns. The Acer, Hitachi and Epson saw a lot of banding and moiré patterns.

In the colour and greyscale tests, the Sharp yet again dominated the scene. The only place where it faltered was the 16 intensity levels, where it missed one colour intensity test. The LCDs in this segment had a problem with streaking and ghosting artefacts, and this was reflected in their poor scores. The BenQ and Sharp showed great resistance to black and white level shifting, which means they can handle high dynamic ranges much better. Not surprisingly, the colour tests saw the Epson give better results, as colours are the strongest points of LCD projection systems. It was the same case with the Hitachi. The Acer performed miserably in the colour segment, with too much colour contamination and absolutely no gradation accuracy in the 16 intensity levels test. The LCDs performed well in the colour accuracy and purity tests, with the Hitachi coming close to the excellence offered by the Sharp in colour reproduction.

Sharp XR-10X

The brilliant screen uniformity was evident in the Sharp, while all the others had some issues with brightness variation and noise across the different areas of the screen. No noticeable flicker was apparent on any of the projectors. The Sharp aced the difficult ANSI Brightness test, the most amazing thing being its ability to distinguish between 5 per cent brightness and contrast variation, which is even beyond the requirements of the test, while being better than any other projector's capability. The BenQ failed the test due to its inability to render finer black levels. The Epson's resolution and focusing problems were apparent in the Defocusing, Blooming and Halo tests.

The Movie Test

The kind of high-grade test material we used for these tests are very unforgiving when it comes to colour inaccuracies. DLP projectors being digital in nature cannot produce accurate colour gradations if their image processor isn't fast enough. Every DLP projector had the previous generation DLP engine; only the Sharp was endowed with a state-of-the-art DDR DLP system with a really fast 3x colour wheel as opposed to the 2x colour wheel the others had.

Also, the new fabrication process of its improved DLP chip endows it with excellent colour accuracy and variation across a larger dynamic range. This was evident in its excellent

Choosing The Right Screen

You may buy a Rs 2 lakh projector, but it will not perform as well if you pair it with a cheap screen. As a rule of thumb, 20 to 30 per cent of the projector budget should be allocated for buying a screen to go along with it. Anything less than 20 per cent of the total projector cost means that you are not getting the full image quality you paid for in the first place.

A projector screen, can, in fact, compensate for the shortcomings of your projector. For example, a high-gain screen will compensate for the low brightness rating of your projector, while conversely, a grey screen with high contrast will enhance the contrast ratio and enable your LCD projector to render purer blacks. Basically, "gain" is the measurement of reflectivity of the projection surface. It is denoted by a unit known as the gain number, which is the ratio of the light reflected by a screen in comparison to the reflectivity of a standard reference white magnesium oxide board. So a gain rating of 1.3 reflects 30 per cent more light. Grey screens used for better contrast have a negative screen gain quotient like .80 gain rating which translates into 20 per cent reduction in gain, but at the added benefit of better black reproduction.

Gain isn't always good, however, as it sees a lot of problems with directionality. Since it has a reflective nature, it significantly reduces the viewing angle of the screen. The problem is more severe on very high-gain screens with hot spots apparent at the middle of the screen. This makes the screen appear brighter in the centre as opposed to the darker corners.

Quite simply, if your projector is lacking in brightness, opt for a high-gain screen. If it is plagued with low contrast issues, buy a a high-contrast screen. Also, high ambient light conditions warrant a high-gain screen, because a high-gain screen will reject omnidirectional ambient light and only reflect the direct projector light.

LCDs generally have poor contrast ratios and suffer at black reproduction; they are therefore well mated with a grey screen with low gain. A DLP, which traditionally has better brightness and contrast, on the other hand, can use both screens effectively.